

The cardinality of a set containing the pairwise sums of a fixed number of integers

Wouter van Doorn

Abstract

Revisiting a 50-year-old estimate of Choi, Erdős and Szemerédi, we show that if $A \subseteq \{1, 2, \dots, 2n\}$ satisfies $|A| \geq n + 1.2 \cdot 10^8$, then there exist five distinct integers whose pairwise sums are all contained in A . In order to guarantee pairwise sums of three or four integers instead, we show that one can replace the constant $1.2 \cdot 10^8$ by 1 or 3 respectively, which are both optimal.

1 Introduction

For positive integers n and $k \geq 3$, let $g_k(n)$ be the smallest integer such that for all sets $A \subseteq \{1, 2, \dots, 2n\}$ with $|A| \geq n + g_k(n)$, one can find distinct integers $b_1, b_2, \dots, b_k$ (not necessarily contained in A) with $b_i + b_j \in A$ for $1 \leq i < j \leq k$. Choi, Erdős and Szemerédi introduced this function in [1], and proved many bounds on it (which were already announced three years prior in [2, p. 83]). For example, they proved

$$\begin{aligned} g_3(n) &\leq 2, \\ g_4(n) &= O(1), \\ g_5(n) &= O(\log(n)), \\ g_6(n) &\asymp \sqrt{n}. \end{aligned}$$

They did not necessarily attempt to optimize all the implied constants however, and estimating the value of $g_k(n)$ is now listed as Erdős Problem #866 at [3]. In this paper we revisit the definition of g_k , and prove that for all $n \geq 3$ we have $g_3(n) = 1$, $g_4(n) = 3$ and $g_5(n) < 1.2 \cdot 10^8$.

2 Declaration of AI usage

In this research ChatGPT (model: GPT-5.3 Instant) was initially used for brainstorming ideas. Most notably, it autonomously came up with the proof of Theorem 3 that we outline below. We further used the automated theorem proving tool Aristotle from Harmonic [4] to obtain Lean formalizations of essentially all results in this paper. More precisely, every stated result with a blue checkmark symbol has been successfully formalized, and the Lean file containing these formal proofs can be found at the author's GitHub page [5]. Particularly noteworthy here is that, in the process of formalizing the result that $h_4(n) \leq 3166$ for all n ,¹ Aristotle managed to come up with further improvements, eventually showing $h_4(n) \leq 2270$.

¹See Section 3 for a definition of h , and Section 7 for a proof sketch of the claimed upper bound.

3 Clarifying the main definition

Regarding the definition of $g_k(n)$, in all theorem statements in [1] the elements in $A \subseteq \{1, 2, \dots, 2n\}$ are referred to as *positive* integers, whereas the b_i (whose pairwise sums need to be contained in A) are just referred to as integers. It is never acknowledged however, that with this definition it is not actually required that the b_i are positive themselves as well. Of course, if both $b_1 \leq 0$ and $b_2 \leq 0$ (say), then $b_1 + b_2 \leq 0 \notin A$, from which we deduce that at most one of the b_i can be non-positive. But allowing (at most one of) the b_i to be non-positive does actually matter! For example, it is claimed that the set

$$A := \{1, 3, \dots, 2n - 1\} \cup \{2, 4, \dots, 2^{\lfloor \log_2 2n \rfloor}\} \quad (1)$$

is an example for which

one cannot choose $b_1, \dots, b_5$ such that $b_i + b_j$ ($1 \leq i < j \leq 5$) are all in A . [1, p. 40]

In particular, this would give the lower bound $g_5(n) > \log_2 n$. This is incorrect as stated, however. With

$$b_1 := -1, b_2 := 2, b_3 := 3, b_4 := 5, b_5 := 6,$$

one can verify that all pairwise sums do belong to A whenever $n \geq 6$, contradicting the above quote.

We are therefore led to define the function $h_k(n)$ as the smallest integer such that for all sets $A \subseteq \{1, 2, \dots, 2n\}$ with $|A| \geq n + h_k(n)$, one can find k distinct *positive* integers whose $\binom{k}{2}$ pairwise sums are contained in A . With this definition, later on we will see that we do recover the lower bound $h_5(n) > \log_2 n$, but that $g_k(n)$ and $h_k(n)$ are in general not the same function. That being said, the fact that (even if we only assume $b_i \in \mathbb{Z}$) at most one of the b_i is non-positive, does imply the bounds

$$g_k(n) \leq h_k(n) \leq g_{k+1}(n) \quad (2)$$

for all $k \geq 3$ and all $n \geq 1$.

4 Sums of three or four integers

As we mentioned, the proposed lower bound example for $g_5(n)$ does not work if we do not put further restrictions on the b_i . Similarly, the example

$$A := \{1, 3, \dots, 2n - 1\} \cup \{2\}$$

given in [1] only helps to show $h_3(n) = 2$ for all $n \geq 4$. For $g_3(n)$ on the other hand, we actually claim that it is equal to 1.

✓ **Theorem 1.** *We have $g_3(n) = 1$ for all $n \geq 3$.*

Proof. The lower bound $g_3(n) \geq 1$ follows from taking A to be the set of all odd integers in $\{1, 2, \dots, 2n\}$, as at least one sum $b_i + b_j$ must be even by the pigeonhole principle.

For the corresponding upper bound, let $A \subseteq \{1, 2, \dots, 2n\}$ be any set with $|A| \geq n + 1$, and assume $n \geq 3$. Then, once again by the pigeonhole principle, A contains $\{m, m + 1\}$ for some odd $m \in \{1, 3, \dots, 2n - 1\}$. If $A \setminus \{m\}$ only contains even integers, then

$$\{2, 4, 6\} \subseteq \{2, 4, \dots, 2n\} \subset A,$$

and we can choose

$$b_1 := 0, b_2 := 2, b_3 := 4.$$

On the other hand, if A contains an odd integer $2j + 1 \neq m$, set

$$b_1 := j, b_2 := j + 1, b_3 := m - j.$$

We then need to show that the b_i are distinct and that their pairwise sums are contained in A . First of all, it is clear that $b_1 \neq b_2$. Secondly, $b_1 \neq b_3$ as we would otherwise have $m = 2j$ contradicting the fact that m is odd. Thirdly, $b_2 \neq b_3$ by the assumption $2j + 1 \neq m$. And finally, the set of pairwise sums is $\{2j + 1, m, m + 1\}$ which is indeed a subset of A . $\square$

✓ **Remark.** *The examples $A := \{1, 2\}$ and $A := \{2, 3, 4\}$ show $g_3(1) = g_3(2) = 2$.*

As it turns out (and despite our insistence on allowing the b_i to be negative), for Theorem 1 we do not actually need to use any negative integers.

✓ **Theorem 2.** *If $n \geq 3$ and $A \subseteq \{1, 2, \dots, 2n\}$ is a set with $|A| \geq n + 1$, then integers $0 \leq b_1 < b_2 < b_3$ exist whose pairwise sums are contained in A .*

Proof. We will prove it by induction; the interested reader is invited to verify it for $n = 3$. We then let

$$A = \{a_1, a_2, \dots, a_r\} \subseteq \{1, 2, \dots, 2n\}$$

be any set with at least $n + 1$ elements, and assume that the theorem is proven for all sets $A' \subseteq \{1, 2, \dots, 2n - 2\}$ with at least n elements. Now, if $1 \notin A$, define

$$A' := \{a_1 - 2, a_2 - 2, \dots, a_r - 2\} \setminus \{0\}.$$

Then the induction hypothesis applies to A' , so let b'_1, b'_2, b'_3 be three non-negative integers whose pairwise sums are in A' . Then we can simply choose $b_i := b'_i + 1$, as $b'_i + b'_j \in A'$ implies $b_i + b_j = b'_i + b'_j + 2 \in A$. On the other hand, if $1 \in A$, then there are two options. Either $\{m, m + 1\} \subset A$ for some $m \geq 2$, in which case we can set $b_1 := 0, b_2 := 1, b_3 := m$. Alternatively, if A does not contain any pair of consecutive integers besides possibly $\{1, 2\}$, then A must be exactly equal to $\{1, 2, 4, 6, \dots, 2n\}$. In that case we can set $b_1 := 0, b_2 := 2, b_3 := 4$. $\square$

From $k = 3$ we now move on to $k = 4$.

✓ **Theorem 3.** *We have $g_4(n) = 3$ for all $n \geq 2$.*

Proof. For the lower bound, we choose

$$A := \{1, 3, \dots, 2n-1\} \cup \{2n-2, 2n\}.$$

To see why A does not contain the pairwise sums of any four integers b_1, b_2, b_3, b_4 , we are free to assume that b_1 and b_2 are even, and b_3 and b_4 are both odd. Indeed, if three of them have the same parity, then their three pairwise sums are all even, while A only contains two even integers. Furthermore assume without loss of generality $b_1 < b_2$ and $b_3 < b_4$. Now, because the smallest even integer in A is $2n-2$, we must have that both $b_1 + b_2$ and $b_3 + b_4$ are at least $2n-2$. In particular, b_2 and b_4 are both at least n , which means their sum is at least $2n+1$ (as they cannot both be equal to n), which is not an element of A .

As for the upper bound, let $A \subseteq \{1, 2, \dots, 2n\}$ be any set with $|A| \geq n+3$. Since the result is vacuous for $n = 2$, we may assume $n \geq 3$. The pigeonhole principle then implies that there exist three distinct even integers a_1, a_2, a_3 such that A contains $\{a_i, 2n+1-a_i\}$ for all $1 \leq i \leq 3$. We now set

$$\begin{aligned} b_1 &:= \frac{a_1 + a_2 - a_3}{2}, \\ b_2 &:= \frac{a_1 + a_3 - a_2}{2}, \\ b_3 &:= \frac{a_2 + a_3 - a_1}{2}, \\ b_4 &:= \frac{4n+2 - a_1 - a_2 - a_3}{2}. \end{aligned}$$

These integers are defined so that we have

$$\begin{aligned} b_1 + b_2 &= a_1, \\ b_1 + b_3 &= a_2, \\ b_1 + b_4 &= 2n+1-a_3, \\ b_2 + b_3 &= a_3, \\ b_2 + b_4 &= 2n+1-a_2, \\ b_3 + b_4 &= 2n+1-a_1. \end{aligned}$$

In particular, $b_i + b_j \in A$ for all $1 \leq i < j \leq 4$. Finally, b_1, b_2, b_3 are all distinct by the assumption that a_1, a_2, a_3 are distinct as well, while $b_4 \notin \{b_1, b_2, b_3\}$ by the fact that b_4 has the opposite parity. $\square$

5 Lower bounds for sums of five integers

Let us first formally verify that the example Choi, Erdős and Szemerédi propose, does work to provide a lower bound on $h_5(n)$.

✓ **Theorem 4** (Choi-Erdős-Szemerédi). *We have $h_5(n) > \log_2 n$ for all $n \in \mathbb{N}$.*

Proof. With A as in (1), we note that we indeed have $|A| > n + \log_2 n$ for all n . Hence, with $b_1, b_2, \dots, b_5$ five arbitrary distinct positive integers, it is sufficient to show that at least one of the sums $b_i + b_j$ (with i, j distinct) is not contained in A . Now, by the pigeonhole principle at least three of the b_i have the same parity.

Let us first assume that there are three b_i that are all even, so that without loss of generality we may write

$$b_1 = 2^{l_1} m_1, b_2 = 2^{l_2} m_2, b_3 = 2^{l_3} m_3$$

with m_1, m_2 , and m_3 odd, and $1 \leq l_1 \leq l_2 \leq l_3$. Now, if one of the latter two inequalities is strict, then

$$b_1 + b_3 = 2^{l_1} (m_1 + 2^{l_3-l_1} m_3)$$

is an even integer which is not a power of two, and hence not in A . If $l_1 = l_2 = l_3$ on the other hand, then two of the m_i (say m_1 and m_2) are congruent modulo 4, in which case

$$b_1 + b_2 = 2^{l_1} (m_1 + m_2)$$

is an even integer which is not a power of two, as $m_1 + m_2 \equiv 2 \pmod{4}$ with $m_1 + m_2 > 2$.

Since the latter argument still works when $l_1 = l_2 = l_3 = 0$, this also deals with the case where at least three of the b_i are odd, finishing the proof. $\square$

Let us now return to g_k , for which we have the following, seemingly modest, lower bound for $k = 5$.

✓ **Theorem 5.** *We have $g_5(n) \geq 4$ for all $n \geq 3$.*

Proof. More precisely, we will prove that for the set

$$A := \{1, 3, \dots, 2n-1\} \cup \{2n-4, 2n-2, 2n\}, \quad (3)$$

there are no five distinct integers whose pairwise sums are all in A . So let b_1, b_2, b_3, b_4, b_5 be any distinct integers and assume by contradiction that A does contain all pairwise sums. If at least four of them have the same parity, then there are at least five distinct even sums, while A only contains three even integers. We may therefore assume without loss of generality

$$b_1 \equiv b_2 \equiv b_3 \not\equiv b_4 \equiv b_5 \pmod{2}$$

with

$$b_1 < b_2 < b_3 \quad \text{and} \quad b_4 < b_5.$$

Given the fact that $2n - 4, 2n - 2$ and $2n$ are the only even integers in A , we then deduce

$$\begin{aligned} b_1 + b_2 &= 2n - 4, \\ b_1 + b_3 &= 2n - 2, \\ b_2 + b_3 &= 2n, \\ b_4 + b_5 &\geq 2n - 4. \end{aligned}$$

Solving the first three equations gives

$$b_1 = n - 3, b_2 = n - 1, b_3 = n + 1,$$

while the final inequality implies $b_5 \geq n - 1$, so that we actually get $b_5 \geq n$ (as we already have $b_2 = n - 1$). But then $b_3 + b_5 \geq 2n + 1 \notin A$. $\square$

The example set as defined in (3) is optimal in the following narrow sense.

✓ **Theorem 6.** *If A is a set with*

$$\{1, 3, \dots, 2n - 1\} \subset A \subseteq \{1, 2, \dots, 2n\}$$

and $|A| \geq n + 4$, then distinct integers $b_1, b_2, \dots, b_5$ exist such that $b_i + b_j \in A$ for all $1 \leq i < j \leq 5$.

As neither the statement nor the proof of Theorem 6 is particularly illuminating (especially in light of the main result in the next section), we have moved the proof to Appendix A.

6 A uniform upper bound for sums of five integers

Using arguments that are essentially borrowed from [1], this section is aimed at proving that $g_5(n)$ is uniformly bounded. We first need a preliminary result that upper bounds $g_k(n)$ if we restrict to sets of even integers. To be able to state it, for $x \geq 1$ we define

$$f_3(x) := \sqrt{\frac{x}{2}} + \left(\frac{x}{2}\right)^{1/4} + \frac{1}{2} \quad (4)$$

and, for $k \geq 4$, we further define

$$f_k(x) := \sqrt{2xf_{k-1}(x) + \frac{1}{4}} + \frac{1}{2}. \quad (5)$$

We then have the following slightly more optimized version of the corollary to Lemma A in [1].

✓ **Lemma 7.** *If $A_0 = \{2 \leq a_1 < a_2 < \dots < a_r\} \subset \mathbb{N}$ is a set of even integers for which the inequality $r \geq f_k(a_r - a_1)$ holds for some $k \geq 3$, then distinct integers $b_1, b_2, \dots, b_k$ exist such that A_0 contains all $\binom{k}{2}$ pairwise sums $b_i + b_j$.*

Proof. We will prove by induction that $r \geq f_k(a_r - a_1)$ implies the existence of integers $c_1, c_2, \dots, c_k$ with $c_2, c_3, \dots, c_k$ non-zero and pairwise distinct, such that A_0 contains all subset sums containing c_1 . That is, for any $S \subseteq \{1, 2, \dots, k\}$ with $1 \in S$, we have $\sum_{i \in S} c_i \in A_0$. If such c_i indeed exist, then the lemma follows by taking $b_1 := \frac{1}{2}c_1$ and $b_i := \frac{1}{2}c_1 + c_i$ for $2 \leq i \leq k$.

We first need to prove it for $k = 3$. In this case, if a positive integer m and indices $i_1 < i_2, j_1 < j_2$ exist with

$$a_{j_1} - a_{i_1} = a_{j_2} - a_{i_2} = m,$$

then one can choose

$$c_1 := a_{i_2}, c_2 := a_{i_1} - a_{i_2}, c_3 := m,$$

where we have $c_2 < 0 < c_3$. However, if such an m and indices do not exist, then $\{a_1/2, a_2/2, \dots, a_r/2\}$ forms a so-called Sidon set, for which the bound

$$r < \sqrt{\frac{a_r}{2} - \frac{a_1}{2}} + \left(\frac{a_r}{2} - \frac{a_1}{2}\right)^{1/4} + \frac{1}{2} = f_3(a_r - a_1)$$

was proved in [6]. Now let $k > 3$ and assume that the lemma holds for $k - 1$.

There are a total of $\frac{1}{2}r(r-1)$ differences $a_j - a_i$ in the interval $[2, a_r - a_1]$, and all of them are even. Hence, by the pigeonhole principle, if

$$\frac{1}{2}r(r-1) \geq (a_r - a_1)f_{k-1}(a_r - a_1), \quad (6)$$

then there are positive integers m and $s \geq f_{k-1}(a_r - a_1)$ such that m can be written in s disjoint ways as a difference

$$a_{j_1} - a_{i_1} = a_{j_2} - a_{i_2} = \dots = a_{j_s} - a_{i_s} = m.$$

And indeed, with definition (5) in mind, inequality (6) is precisely equivalent to $r \geq f_k(a_r - a_1)$ by the quadratic formula. For simplicity, write

$$S := \{a_{i_1}, a_{i_2}, \dots, a_{i_s}\} \quad \text{and} \quad T := \{a_{j_1}, a_{j_2}, \dots, a_{j_s}\}.$$

Now we apply the induction hypothesis to S , to obtain $c_1, c_2, \dots, c_{k-1}$ for which all the subset sums containing c_1 belong to S . We then set $c_k := m$ and note that c_k is distinct from the integers $c_2, c_3, \dots, c_{k-1}$, because otherwise $c_1 + m \in S \cap T$, contradicting the fact that S and T are disjoint. With $c_k = m$ included, we deduce that all the subset sums containing c_1 now belong to $S \cup T \subseteq A_0$, as desired. $\square$

We are now ready to prove our main theorem, by noting that from the definitions in (4) and (5), one can verify by induction that for all $k \geq 3$ the inequalities²

²In fact, for the first inequality the explicit estimate of

$$f_k(x) \leq 2^{1-\frac{3}{2^{k-2}}} x^{1-\frac{1}{2^{k-2}}} + 2x^{1-\frac{3}{2^{k-1}}}$$

is possible.

$$f_k(x) \leq (2 + o(1))x^{1 - \frac{1}{2^{k-2}}} \quad (7)$$

and

$$2f_k(x) > f_k(2x) \quad (8)$$

hold for all $x \geq 1$.

✓ **Theorem 8.** *We have $g_5(n) < 1.2 \cdot 10^8$ for all $n \in \mathbb{N}$.*

Proof. Let C be the smallest integer for which the inequality

$$\frac{x}{12} + \frac{C}{2} - 2 > f_5(x) \quad (9)$$

holds for all $x \geq 1$. We remark that such a C certainly exists, as the left-hand side grows linearly, while the right-hand side has sublinear growth by the upper bound (7). In fact, calculating the derivative of $\frac{x}{12} - f_5(x)$ and setting it equal to 0 gives

$$C = 113,591,719 < 1.2 \cdot 10^8.$$

Now let $A \subseteq \{1, 2, \dots, 2n\}$ be any set with $|A| \geq n + C$, let $A_0 \subset A$ be the subset of all even integers in A , and write $|A_0| = t + C$ for some $t \geq 0$. If $t \geq \frac{n}{6}$, then

$$|A_0| \geq \frac{n}{6} + C > f_5(2n)$$

for all $n \geq 1$ by multiplying inequality (9) by 2 and applying (8), which implies we are done by Lemma 7. We may therefore assume $t < \frac{n}{6}$ for the rest of this proof.

Now, if there are at most 4 even elements of A contained in the closed interval $[6t + 2, 2n - 6t - 2]$, then $t > 0$ and there are either at least

$$r := \left\lceil \frac{|A_0| - 4}{2} \right\rceil = \left\lceil \frac{t}{2} + \frac{C}{2} - 2 \right\rceil$$

even elements of A in the interval $[2, 6t]$, or at least r even elements of A in $[2n - 6t, 2n]$. In either situation, with $x := 6t$ we have an interval of length at most x with at least r even elements. Lemma 7 then finishes the proof again, as

$$r \geq \frac{t}{2} + \frac{C}{2} - 2 = \frac{x}{12} + \frac{C}{2} - 2 > f_5(x),$$

by inequality (9).

Hence, without loss of generality we may assume that there are at least 5 even elements of A contained in the interval $[6t + 2, 2n - 6t - 2]$. In particular, we may assume the existence of three even integers a_1, a_2, a_3 with

$$6t + 2 \leq a_1 < a_2 < a_3 \leq n$$

or three even integers a'_1, a'_2, a'_3 with

$$n \leq a'_1 < a'_2 < a'_3 \leq 2n - 6t - 2.$$

Similar to the proof of Theorem 3, we then define

$$\begin{aligned} b_1 &:= \frac{a_1 + a_2 - a_3}{2}, \\ b_2 &:= \frac{a_1 + a_3 - a_2}{2}, \\ b_3 &:= \frac{a_2 + a_3 - a_1}{2} \end{aligned}$$

in the first case, or

$$\begin{aligned} b'_1 &:= \frac{a'_1 + a'_2 - a'_3}{2}, \\ b'_2 &:= \frac{a'_1 + a'_3 - a'_2}{2}, \\ b'_3 &:= \frac{a'_2 + a'_3 - a'_1}{2} \end{aligned}$$

in the second case. As before,

$$\begin{aligned} b_1 &< b_2 < b_3, \\ b_1 &\equiv b_2 \equiv b_3 \pmod{2}, \\ \{b_1 + b_2, b_1 + b_3, b_2 + b_3\} &\subset A. \end{aligned}$$

And similarly for b'_1, b'_2, b'_3 .

In the first case we consider the set of all integers p, q with $b_1 \not\equiv p \equiv q \pmod{2}$ such that $p + q = a_3$ and

$$\frac{a_3}{2} - 6t - 2 \leq p < q \leq \frac{a_3}{2} + 6t + 2,$$

and in the second case we consider the set of all integers p, q with $b'_1 \not\equiv p \equiv q \pmod{2}$ such that $p + q = a'_1$ and

$$\frac{a'_1}{2} - 6t - 2 \leq p < q \leq \frac{a'_1}{2} + 6t + 2.$$

In both cases there are at least $3t + 1$ such pairs (p, q) , while we claim that all

pairwise sums are contained in $[1, 2n]$. Indeed $p + q \in \{a_3, a'_1\} \subset [1, 2n]$, while

$$\begin{aligned}
p + b_1 &\geq \frac{a_3}{2} - 6t - 2 + \frac{a_1 + a_2 - a_3}{2} \\
&\geq -6t - 2 + \frac{(6t + 2) + (6t + 4)}{2} \\
&= 1, \\
q + b_3 &\leq \frac{a_3}{2} + 6t + 2 + \frac{a_2 + a_3 - a_1}{2} \\
&\leq \frac{n}{2} + 6t + 2 + \frac{(n - 2) + n - (6t + 2)}{2} \\
&= \frac{3n}{2} + 3t \\
&< 2n
\end{aligned}$$

in the first case, and

$$\begin{aligned}
p + b'_1 &\geq \frac{a'_1}{2} - 6t - 2 + \frac{a'_1 + a'_2 - a'_3}{2} \\
&\geq \frac{n}{2} - 6t - 2 + \frac{n + (n + 2) - (2n - 6t - 2)}{2} \\
&= \frac{n}{2} - 3t \\
&> 0, \\
q + b'_3 &\leq \frac{a'_1}{2} + 6t + 2 + \frac{a'_2 + a'_3 - a'_1}{2} \\
&\leq 6t + 2 + \frac{(2n - 6t - 4) + (2n - 6t - 2)}{2} \\
&= 2n - 1
\end{aligned}$$

in the second case.

Now, A misses at most $|A_0| - C = t$ odd integers in the interval $[1, 2n]$, while in either case we have at least $3t + 1$ pairs (p, q) to work with. Hence, in the first case there must be a pair (p, q) such that

$$b_1 + p, b_2 + p, b_3 + p, b_1 + q, b_2 + q, b_3 + q$$

are all in A , as each missing odd integer rules out at most three pairs. We may therefore take $b_4 := p, b_5 := q$. The analogous argument replacing b_1, b_2, b_3 by their primed counterparts works for the second case. $\square$

7 Concluding thoughts and remarks

In [1] it was proved that $h_4(n) = O(1)$. For an explicit variant, note that Theorem 8 implies by inequality (2) that we actually have $h_4(n) < 1.2 \cdot 10^8$.

However, a much better bound is possible. For this, we recall that Ruzsa [7, Theorem 4.6] proved that for any set $A \subseteq \{1, 2, \dots, n\}$ with

$$|A| > \sqrt{n} + 4n^{1/4} + 11$$

there are four distinct integers $a_1 < a_2 < a_3 < a_4$ in A for which $a_1 + a_4 = a_2 + a_3$. Sets for which no such a_i exist are called weak Sidon sets, and note the difference with the definition of a Sidon set, where even $a_2 = a_3$ is not allowed. In any case, with four such integers, we can define

$$c_1 := a_1, c_2 := a_2 - a_1, c_3 := a_3 - a_1$$

in the proof of the base case of Lemma 7, to obtain positive c_i and hence positive b_i . With the definition in (4) appropriately altered to

$$F_3(x) := \sqrt{\frac{x}{2}} + 4\left(\frac{x}{2}\right)^{1/4} + 11$$

and $F_4(x)$ analogously defined according to (5), the rest of the proof of Lemma 7 then still goes through the exact same way.

To use this to find an upper bound for $h_4(n)$, we sketch an argument similar to the proof of Theorem 8; let $A \subseteq \{1, 2, \dots, 2n\}$ be a set with $|A| \geq n + C$ for some constant C that we have yet to determine, let $A_0 \subset A$ be the subset of all even integers in A , and write $|A_0| = t + C$ for some $t \geq 0$. If there exists an even element

$$2m \in [8t + 6, 2n - 8t - 6] \cap A_0,$$

then first choose $b_1 := m - 1$, $b_2 := m + 1$. Secondly, choose $b_3 \in [m - 4t - 2, m - 2]$ and $b_4 := 2m - b_3$ with $b_3 \equiv b_4 \not\equiv b_1 \pmod{2}$ and such that

$$b_1 + b_3, b_1 + b_4, b_2 + b_3, b_2 + b_4$$

are all elements of A . The latter is possible as there are at least $2t + 1$ possible choices for b_3 , while every odd integer not contained in A eliminates at most two possibilities, and A misses at most t odd integers.

On the other hand, if $[8t + 6, 2n - 8t - 6] \cap A_0$ is empty, then we have at least $\lceil \frac{t}{2} + \frac{C}{2} \rceil$ even elements in an interval of length $x := 8t + 6$. Hence, choosing C large enough so that

$$\frac{x - 6}{16} + \frac{C}{2} > F_4(x)$$

for all $x \geq 1$ would finish the proof, and one can verify that $C = 3166$ does the job. We thereby find $h_4(n) \leq 3166$ for all n , although the formalization that is available at [5] actually provides a slightly better bound, as explained in Section 2. Further improvements are plausible by e.g. applying stronger bounds on weak Sidon sets.

Turning to $k = 5$, despite the rather large constant in Theorem 8, it could be the case that $g_5(n) \leq 5$ holds for all n . We have tried to look for counterexamples up to $n = 15$, but none were found. In fact, we cannot even exclude the possibility that $g_5(n) \leq 4$ for all large enough n , which would be optimal by Theorem 5. We note however that e.g. $A = \{1, 2, 4, 5, 6, \dots, 10\}$ does serve as an example that $g_5(n) \geq 5$ is at least possible for some small n .

As for $k = 6$, let $\{a_1, a_2, \dots, a_r\} \subseteq \{1, 2, \dots, \lfloor n/2 \rfloor\}$ be a Sidon set, and recall that such sets exist with $r = \left(\frac{1}{\sqrt{2}} + o(1)\right) \sqrt{n}$ (see e.g. [8]). Now define

$$A := \{1, 3, \dots, 2n-1\} \cup \{4a_1-2, 4a_2-2, \dots, 4a_r-2\}.$$

In [1] this set A is used to prove the lower bound

$$h_6(n) \geq \left(\frac{1}{\sqrt{2}} - o(1)\right) \sqrt{n},$$

but it works just as well as a lower bound for $g_6(n)$. As they also prove an upper bound with the same order of growth, we have

$$h_k(n) = O(g_k(n)) \tag{10}$$

for $k \in \{3, 4, 6\}$, which one can compare with inequality (2). However, by Theorem 4 and Theorem 8, inequality (10) does of course not hold for $k = 5$.

Finally, similar to how inequality (7) can be shown, for larger values of k one can combine the aforementioned bound on weak Sidon sets with the definition in (5) to prove the following general upper bound, which is a slight improvement over [1, Theorem 5].

✓ **Theorem 9.** *For all $k \geq 3$ there exists an N such that for all $n \geq N$ we have*

$$g_k(n) \leq h_k(n) < 4n^{1-\frac{1}{2^{k-2}}}.$$

Appendix A. Proof of Theorem 6

In this appendix we record the proof of Theorem 6, for completeness' sake.

Proof of Theorem 6. Let $2 \leq a_1 < a_2 < a_3 < a_4 \leq 2n$ be four even integers in A (which exist by the assumption $|A| \geq n+4$), and first assume $a_2 + a_3 \leq 2n$. We then define

$$\begin{aligned} b_1 &:= \frac{a_1 + a_2 - a_3}{2}, \\ b_2 &:= \frac{a_1 + a_3 - a_2}{2}, \\ b_3 &:= \frac{a_2 + a_3 - a_1}{2}, \\ b_4 &:= 1 - b_1, \\ b_5 &:= a_3 - b_4. \end{aligned}$$

We need to check that these b_i are distinct and that the 10 pairwise sums are in A . To see why the b_i are distinct, it is quickly verified that

$$b_1 < b_2 < b_3 \quad (11)$$

and

$$b_1 \equiv b_2 \equiv b_3 \not\equiv b_4 \equiv b_5 \pmod{2}. \quad (12)$$

Hence, to prove distinctness it suffices to show

$$b_4 < b_5. \quad (13)$$

And indeed,

$$b_5 - b_4 = a_3 - 2b_4 = a_3 + 2b_1 - 2 = a_1 + a_2 - 2 > 0.$$

To prove that A contains all 10 pairwise sums, one can quickly verify that the four sums $b_1 + b_2$, $b_1 + b_3$, $b_2 + b_3$ and $b_4 + b_5$ are all in $\{a_1, a_2, a_3\} \subset A$. Furthermore, equation (12) implies that the other six pairwise sums are odd. As A contains by assumption all odd integers, by equations (11) and (13) it is sufficient to show the inequalities

$$1 \leq b_1 + b_4 < b_3 + b_5 \leq 2n - 1. \quad (14)$$

The first inequality is immediate by the definition of b_4 , while for the second inequality we calculate

$$b_3 + b_5 = b_3 + a_3 + b_1 - 1 = a_2 + a_3 - 1 \leq 2n - 1,$$

where the final inequality uses the assumption $a_2 + a_3 \leq 2n$.

The second case we have to deal with is $a_2 + a_3 > 2n$ and $a_3 < 2n - 2$, in which case we define

$$\begin{aligned} b_1 &:= \frac{a_2 + a_3 - a_4}{2}, \\ b_2 &:= \frac{a_2 + a_4 - a_3}{2}, \\ b_3 &:= \frac{a_3 + a_4 - a_2}{2}, \\ b_4 &:= a_2 + b_3 - (2n - 1), \\ b_5 &:= (2n - 1) - b_3. \end{aligned}$$

Analogously, we now have

$$\{b_1 + b_2, b_1 + b_3, b_2 + b_3, b_4 + b_5\} = \{a_2, a_3, a_4\} \subset A,$$

while both equation (11) and (12) are still quickly verified. It is therefore similarly sufficient to show inequalities (13) and (14). And indeed:

$$\begin{aligned} b_5 - b_4 &= (4n - 2) - a_2 - 2b_3 = (4n - 2) - a_3 - a_4 > 0, \\ b_1 + b_4 &= b_1 + a_2 + b_3 - (2n - 1) = a_2 + a_3 - (2n - 1) \geq 1, \\ b_3 + b_5 &= 2n - 1. \end{aligned}$$

The final case is where $a_2 + a_3 > 2n$ and $a_3 = 2n - 2$. We then define

$$\begin{aligned} b_1 &:= \frac{a_2 + a_3 - a_4}{2}, \\ b_2 &:= \frac{a_2 + a_4 - a_3}{2}, \\ b_3 &:= \frac{a_3 + a_4 - a_2}{2}, \\ b_4 &:= a_1 + b_3 - (2n - 1), \\ b_5 &:= (2n - 1) - b_3. \end{aligned}$$

Once again, verifying inequalities (13) and (14) is sufficient:

$$\begin{aligned} b_5 - b_4 &= (4n - 2) - a_1 - 2b_3 = (4n - 2) - a_1 + a_2 - a_3 - a_4 = a_2 - a_1 > 0, \\ b_1 + b_4 &= b_1 + a_1 + b_3 - (2n - 1) = a_1 + a_3 - (2n - 1) = a_1 - 1 \geq 1, \\ b_3 + b_5 &= 2n - 1. \end{aligned} \quad \square$$

References

- [1] S.L.G. Choi, P. Erdős, E. Szemerédi, *Some additive and multiplicative problems in number theory*. Acta Arithmetica, vol. 27, 37–50, 1975.
- [2] P. Erdős, *Extremal problems in number theory*. Proceedings of the 1972 Number Theory Conference (Univ. Colorado, Boulder, Colo.), 80–86, 1972.
- [3] T. F. Bloom, Erdős Problem #866, <https://www.erdosproblems.com>, accessed 2026-04-04.
- [4] T. Achim et al., *Aristotle: IMO-level Automated Theorem Proving*, arXiv:2510.01346 [cs.AI], 2025.
- [5] W. van Doorn, *ErdosProblem866.lean*, GitHub repository, <https://github.com/Woett/Lean-files/blob/main/ErdosProblem866.lean>.
- [6] K. O'Bryant, *On the Size of Finite Sidon Sets*. Ukrainian Mathematical Journal, Volume 76, 1352–1368, 2024.
- [7] I. Z. Ruzsa, *Solving a linear equation in a set of integers I*. Acta Arithmetica, Volume 65, 259–282, 1993.
- [8] K. O'Bryant, *A complete annotated bibliography of work related to Sidon sequences*. The Electronic Journal of Combinatorics, vol. 11, 2004.